



LUCAS W. DAVIS
PAUL GERTLER
NEIL THOMPSON
ERNESTO RODRIGUEZ

Evaluating the Hawthorne Experiments

Tough Times

It is the morning of October 25th, 2020 and Julian Collins is reflecting over the manufacturing strategy used by Fresh Taste Inc. (FTI). As Director of Operations, he is worried about the Omaha, Nebraska plant – FTI’s biggest plant. Much of the plant’s equipment is becoming obsolete; in addition, decreasing profits have affected the company’s ability to make the large capital investments that are necessary to modernize the plant. Julian starts to draft a proposal for increasing productivity levels. Labor expenses are a big share of the company’s per-unit costs, and if he can find a way to lower these costs the Omaha plant would return to profitability.

FTI is the world leader in meat-based packaged products. Over the last few years, however, a number of competing products have entered this market, pushing FTI to improve its productivity levels to keep margins from falling. In addition, the recent recession forced many of FTI’s competitors into bankruptcy. As they emerged from bankruptcy, they were able to renegotiate lower salaries with their workers. Because of these circumstances, as well as FTI’s labor benefit policy, their labor costs are considerably higher than industry averages.

Julian has two options. His first option is to approach the FTI Labor Union (FLU). FLU was upset when FTI cut wages in 2018 in the face of the imminent recession, and is unlikely to accept additional concessions. His second option is to change the work conditions to increase profitability. Evidence from a set of

experiments at the company's Hawthorne plant suggest that improving lighting conditions may be an effective, low-cost way to increase productivity.

A Long History

FTI is a family business incorporated in Omaha, Nebraska in 1923. Its founder, George Leigh, was concerned with his workers' welfare and established one of the first guaranteed annual wage plans. Under this plan, the company promised 52 weeks notice before employment termination and established a weekly minimum number of work hours. FTI was one of the few companies in the meat processing industry that offered year-round employment. In addition, in an innovative approach to employer-labor relationships, George Leigh established a profit-sharing program, enabling employees to receive a share of company profits if production goals were surpassed.

The company enjoyed high profits in the 1950s and 1960s, prompting management to diversify into other products, such as canned beans and soups, and expand operations to neighboring states. Although profits were high in Nebraska thanks to the high quality/cost ratio of livestock, outside Nebraska, FTI's competition was fiercer, causing them to abandon the company's social mission of providing guaranteed annual wage plans for workers.

During the 1970s and 1980s, FTI became the market leader in meat-packaged goods in the United States, specializing in breakfast sausages, ham, beef burgers and hot dogs. FTI profits were a direct result of locally sourced quality meat, streamlined processes and efficient distribution operations. FTI was also one of the early adopters of refrigerated trucks, allowing them to extend the shelf life of its products and reach more markets.

During the 1990s, FTI embarked on training programs to raise the historically low productivity of its employees. After almost a decade, however, productivity levels were still below industry averages. Nevertheless, robust demand growth kept the company from losing money. This all changed in the 2010s as several factors caused the financial outlook of the company to deteriorate considerably.

The main problem for FTI was the recent entry into the market by several new competitors. These new companies had lower labor expenses. Most employed non-unionized employees. In addition, many had automated packaging equipment which reduced the amount of labor required per unit of production. During the same period some long-time competitors filed for bankruptcy and were able to negotiate very low wages with their unions. In a time of recession, people preferred to work for a much lower salaries – sometimes 50% lower – to avoid losing their jobs.

The Hawthorne Experiments

In 2019 the management team at the company's Hawthorne plant performed a series of experiments designed to test the impact of lighting conditions on productivity levels. The experiments took place sporadically during the winter months and involved supplementing the factory's antiquated lighting

system with portable lighting. “Our workers cannot see what they are doing,” explained the plant foreman, “so we wanted to see if better lighting would allow them to work faster and get more done.”

Replacing the plant’s lighting system would have been expensive and the plant managers decided to first perform an experiment. Sporadically during the winter months plant engineers brought in portable generators and free-standing light systems which they were able to rent by the day. The idea was that if this lighting was shown to significantly increase productivity, then the management team would go ahead and raise the capital needed to upgrade the plant’s lighting system.

Three separate departments in the factory participated in the experiment; though workers discovered it was easiest to install the portable lights in department 1 and so they were most often installed there. Daily productivity levels were recorded from all three departments during a 180-day period so there are 540 total observations in the data. As summarized in **Exhibit A**, the initial analysis suggested modest productivity gains. However, disagreements arose among the plant managers about how to interpret the results and nothing more was done.

Desperate times call for desperate measures, and Julian is willing to consider anything that would improve productivity and help FTI survive. Julian believes that if lighting could increase productivity at the Hawthorne plant, it would likely increase productivity in FTI’s other plants as well. Moreover, compared to most of the alternatives, lighting is relatively low cost and can be installed quickly. Based on Julian’s calculations, if lighting could increase productivity by more than 4 units per day then the investment would pay for itself. Still, Julian is not 100% convinced and before making a final decision decides to take another look at the results from the experiment.

Exhibit A: Mean Daily Productivity (Units per Day)

	Number of Observations	Mean	Standard Deviation	Skewness	Kurtosis
Normal Lighting	320	452.2	84.4	-.18	.09
Improved Lighting	220	472.4	85.8	.19	.10

Discussion Questions

1. Who is Julian Collins? What is his situation?
2. What were the Hawthorne Experiments?
3. Describe the results reported in Exhibit A.
4. Do you recommend any additional data analysis? What?